

SMART INDIA HACKATHON 2026

FindIt: Multimodal AI Lost & Found Network

Cross-modal vision-language retrieval with spatiotemporal decay, for automated campus recovery.

Smart Automation

AI Microservice

SBERT + CLIP + FastAPI

Firebase + Render

The Campus Lost Item Crisis

PROBLEM ANALYSIS



Fragmented Channels

Unorganized WhatsApp groups, unofficial social handles, and physical paper registers create isolated data silos with no shared record.



Vocabulary Mismatch

Keyword-based search fails when a finder logs a “leather purse” while the claimant searches for “black wallet” — same item, different words.



Manual Admin Overhead

Campus security desks spend excessive hours sorting physical logs and answering the same repetitive student queries.

Privacy & Security Risks

PROBLEM ANALYSIS



Phone Number Exposure

Broadcasting personal contact details on public group chats leads to spam, harassment, and safety concerns for students.



Unverifiable Ownership Claims

Anyone viewing publicly posted photos can attempt to falsely claim a high-value item, with no way to check who actually reported it.



No Spatial Filtering

Distant or stale listings clutter search feeds — a system with no location awareness treats a report from across the city the same as one next door.

FindIt: AI-Powered Recovery

CORE SOLUTION

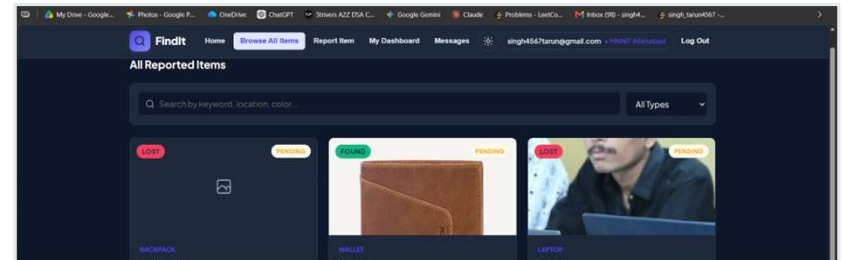
A campus-scoped recovery network that bridges cross-modal descriptions and physical locations — without exposing anyone's personal contact details.

Direct Text-to-Image Matching

Lets a claimant locate a found item using only a written description — no manual photo tagging needed on either side.

Autonomous Score Calibration

Normalizes raw cosine similarities onto a calibrated 0–1 scale, so a displayed match percentage means something consistent.



Real product — Browse view with live status badges

Community & Trust Layer

CORE SOLUTION

Self-Selected Campus Network

At sign-up, students choose their institution from a category list (IIT / NIT / IIIT / Other). Every report and match is scoped to that college — students only ever see activity from their own campus.

Private In-App Messaging

Owners and finders coordinate entirely inside the app via real-time chat — no phone numbers are ever exchanged, closing the exposure risk from the problem slide.

Verified Email Sign-Up

A real verification link is sent on sign-up. Reporting an item and sending messages are both blocked until the email is confirmed — stopping throwaway or non-existent addresses from acting on the platform.

Cross-Modal Vision-Language



SBERT Text Vectors

all-MiniLM-L6-v2 generates 384-dimensional dense semantic vectors, capturing contextual meaning across item titles, categories, and landmarks.



CLIP Visual Projections

OpenAI CLIP (ViT-B/32) maps image pixels and text prompts into a shared 512-dimensional embedding space for zero-shot retrieval.



Dual Score Calibration

Min-max rescaling suppresses noise-floor artifacts (Image: 0.62–0.90 → 0–1; Cross-Modal: 0.15–0.35 → 0–1), producing honest confidence values.

Mathematical Late-Fusion

MATHEMATICAL MODELING

5-Way

Multimodal Fusion Formula

Dynamic weights calibrated across Text, Image, Cross-Modal, Geo, and Temporal signals.

Composite Match Objective

$$S = w^t \cdot S_{\text{text}} + w^i \cdot S_{\text{image}} + w^c \cdot S_{\text{cross}} + w^g \cdot S_{\text{geo}} + w^t \cdot S_{\text{time}}$$

$$S_{\text{geo}} = \exp(-d^2/2\sigma^2) \quad (\sigma = 200\text{m})$$

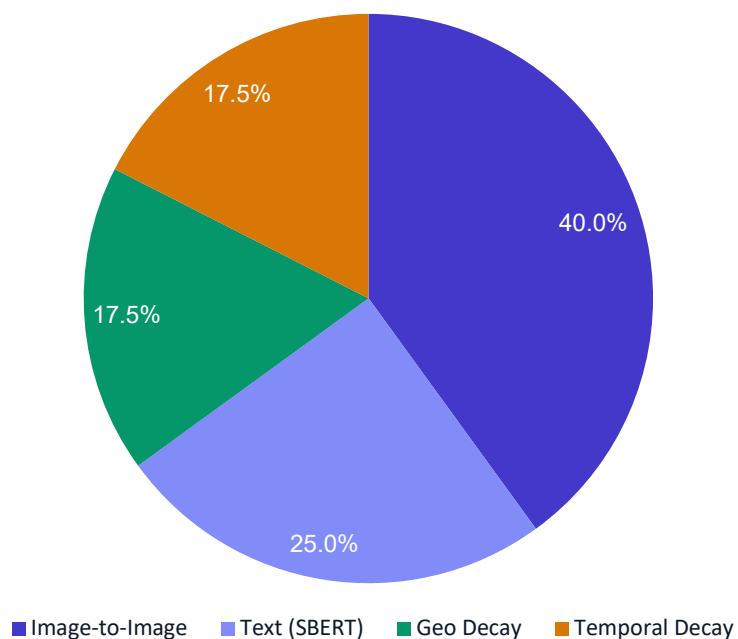
$$S_{\text{time}} = \exp(-\Delta t/\tau) \quad (\tau = 48\text{h})$$

Found timestamps more than 2 hours before the loss are scored zero.

Adaptive Weighting Modes

SCORE FUSION ALGORITHM

Both Photos Present



Single Photo Present

Text 30% • Cross-Modal (photo $\leftrightarrow$ text) 35%
Geo Decay 17.5% • Temporal Decay 17.5%

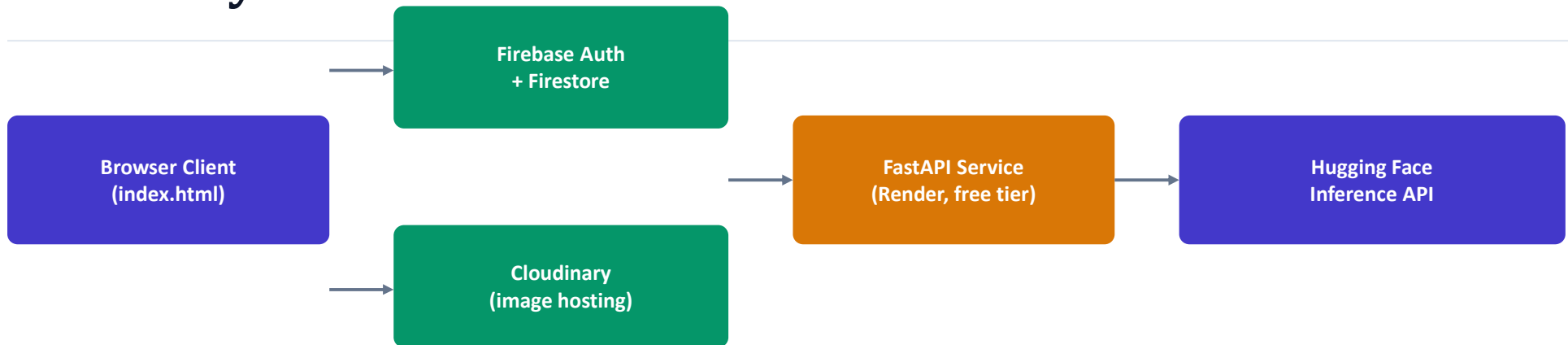
Text-Only Query

Text (SBERT) 60% • Geo Decay 20% • Temporal Decay 20%

Weights re-balance automatically toward whichever signals are actually available for a given pair — a missing photo never breaks the match, it just shifts trust to text and location.

Actual System Architecture

SYSTEM DESIGN



Core app data (auth, items, chat) never touches Render — only the AI matching call does.

Why offload to Hugging Face instead of running PyTorch on Render directly?

Running SBERT + CLIP locally needs real GPU/CPU and RAM — more than a free hosting tier provides. Calling Hugging Face's hosted inference API keeps our own FastAPI service lightweight and deployable for free, at the cost of an extra network hop per request.

Honest trade-off: this is why a request can feel slow — Render's free tier sleeps after ~15 minutes idle, and every match also depends on Hugging Face's own response time.

Trust & Safety Features

SECURITY FRAMEWORK



Mandatory Email Verification

A real verification link must be confirmed before an account can report items or send messages — blocks throwaway or non-existent addresses.



Zero Phone Number Exposure

All coordination happens through in-app real-time chat. A username is shown to other users — never a raw email or phone number.



College-Scoped Visibility

Firestore queries filter every report and match by the reporter's self-selected college, so campuses never see each other's listings.

Known Limitations, Stated Honestly

ENGINEERING HONESTY

We'd rather name these directly than have them discovered live:

College is self-selected, not domain-verified

Sign-up doesn't yet check that a student's email domain matches their claimed institution. A clear, scoped next step.

No ownership challenge-response yet

Handover currently relies on in-app chat and human judgment — there's no built-in “prove you own this” verification step.

ML backend has a silent fallback

If the Hugging Face inference call fails, the engine currently falls back to a placeholder vector rather than surfacing an error — being hardened.

Risk Management Matrix

Risk Factor	Impact	Mitigation Status
Cold-Start Backend Latency	MEDIUM	Known issue on Render's free tier. Planned fix: a keep-alive ping service (e.g. UptimeRobot) — not yet configured.
ML API Failure → Silent Fallback	HIGH	Currently falls back to a placeholder vector without warning. Actively being hardened to fail visibly instead.
Unverified College Claims	MEDIUM	Mitigated partially by category dropdown; “Other” remains free-text with no domain check.
Missing Query Modality	LOW	Handled — fusion weights dynamically re-balance across text-only, single-photo, and dual-photo modes.

Societal & Campus Benefits

VALUE PROPOSITION



Student Welfare

Faster, privacy-safe recovery of lost student IDs, keys, and laptops — without ever broadcasting a personal phone number.



Administrative Efficiency

Replaces tedious physical register logs with searchable, automated attribute indexing, reducing manual sorting at security desks.



Sustainability & E-Waste

Helps prevent premature replacement purchases of functional electronics by improving the odds a lost device finds its way back.

Impact framed directionally — formal measurement (recovery-rate lift, inquiry-volume reduction) is a planned next step, not yet benchmarked.

Scalability & Roadmap

FUTURE SCOPE

Phase 1: Production (Done)

Decoupled FastAPI microservice with live SBERT, CLIP, Firestore, and GPS-aware late-fusion matching, deployed on Render + Firebase Hosting.

Phase 3: Persistent Vector Index

Replace the in-memory found-items list with a real vector database (e.g. pgvector or FAISS) so matching survives server restarts.

Phase 2: College Domain Verification

Require sign-up emails to match a verified institutional domain, closing the self-selection gap named earlier.

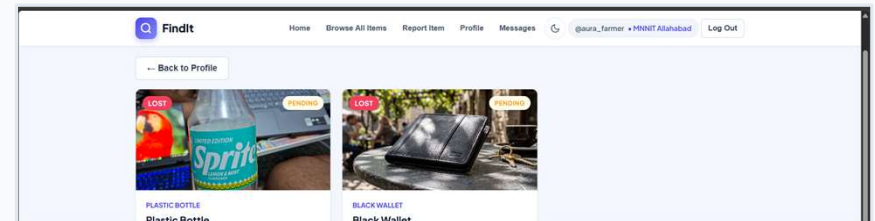
Phase 4: Civic & Multi-Campus Scale

Expand beyond individual colleges toward shared municipal or multi-institution recovery networks.

The Product, Working Today

LIVE VERIFICATION

- Verified email auth, scoped by self-selected campus
- Real AI match scores from a live SBERT + CLIP pipeline
- Pending / Resolved status on every report
- In-app real-time messaging — zero phone numbers shown



Live product — real AI match %, status badges, and @username shown instead of email

SIH 2026

Thank You

Questions, Live Demonstration & Technical Discussion

Live API: lost-and-found-sih-1.onrender.com